Optical Tactile Sensing for Underwater Robotics

A Camera-Based Pipeline for Contact Force, Depth, and Surface Wave Estimation on Soft Robot Skin

Ishan Ahluwalia[1], Annie Yang[2], Yucheng[3], Dr. Cynthia Sung[†]

[1,2,3]University of Pennsylvania [†]Principal Investigator, University of Pennsylvania

github.com/IshanAhluwalia/Optical-Tactile-Sensing-Research

Abstract

We present an optical tactile sensing system for a large-area, curved soft robotic skin designed for underwater deployment. Inspired by vision-based tactile sensors such as GelSight [1] and DenseTact [3], our system prints a dense fractal maze pattern onto a transparent elastomer skin and mounts a fisheye camera internally to observe pattern deformation under contact. A spatiotemporal encoder-decoder network trained on a fully supervised dataset of 1,076 contact locations and approximately 360,000 labeled frames simultaneously estimates contact location, indentation depth, normal force distribution, and surface wave propagation at 30 fps from visual input alone — no force sensors are required at inference. We derive spatially varying material parameters from Hertzian contact mechanics and surface energy theory, enabling physically grounded point cloud label generation across the full skin surface. A baseline ResNet18 regressor achieves 0.52 mm location MAE, 0.38 mm depth MAE, and 55 mN force MAE on fully held-out sessions.

1. Introduction

Tactile sensing is a fundamental capability for robots operating in unstructured or contact-rich environments. Underwater robotics presents a particularly demanding setting: conventional sensor arrays are difficult to waterproof, prone to corrosion, and add significant weight and complexity to the system. Vision-based tactile sensing offers an attractive alternative — a single camera behind a compliant patterned skin can effectively function as a dense tactile array with sub-millimeter spatial resolution [1, 3, 4, 2].

Existing optical tactile sensors are predominantly designed for robotic fingertips: flat or mildly curved surfaces of small area (typically $< 5\,\mathrm{cm}^2$), with controlled internal illumination. Our platform presents qualitatively different challenges. The sensing surface is a *curved oval elastomer* *skin* spanning 225 mm $\times$ 30 mm, mounted on a rigid underwater robot frame. The camera operates under ambient illumination from a fixed internal position, and the system must additionally characterize surface wave propagation resulting from hydrodynamic and contact-induced loading — a sensing modality not addressed by prior fingertip-scale systems.

Our contributions are: (1) a data collection pipeline using a repurposed 3D printer as a precision indentation rig with synchronized force measurement; (2) an analytically derived label generation framework incorporating spatially varying Hertzian contact mechanics and elastic surface tension; (3) a spatiotemporal encoder-decoder architecture with masked multi-head supervision for simultaneous static and dynamic tactile estimation; and (4) a fully collected dataset of 1,076 contact locations with ground truth force, depth, and surface deformation fields.

2. Related Work

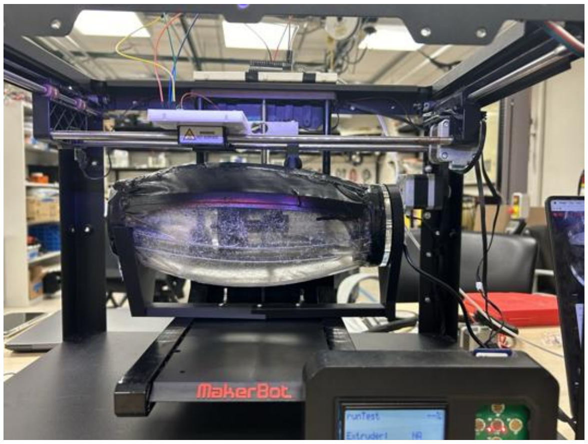
Vision-based tactile sensing. GelSight [1] established the core paradigm of embedding a camera behind a reflective patterned elastomer and using photometric stereo to recover surface geometry. GelSight2 [2] and DIGIT [4] miniaturized this approach for robotic manipulation. DenseTact [3] introduced dome-shaped geometry for multi-directional contact estimation. Tactile-GCN [5] applies graph neural networks to tactile point cloud data. None of these systems addresses large-area curved surfaces or dynamic wave sensing.

Deformable body contact mechanics. Hertz contact theory [6] provides the foundational force-displacement relationship for elastic spherical indentation. Johnson-Kendall-Roberts (JKR) [7] and Derjaguin-Muller-Toporov (DMT) [8] theories extend this to include adhesion effects relevant to soft elastomers. Style et al. [9] demonstrated that elastic surface tension γ becomes significant for soft materials (shear modulus $G < 100\,\text{kPa}$), modifying contact radius and deformation field at small scales.

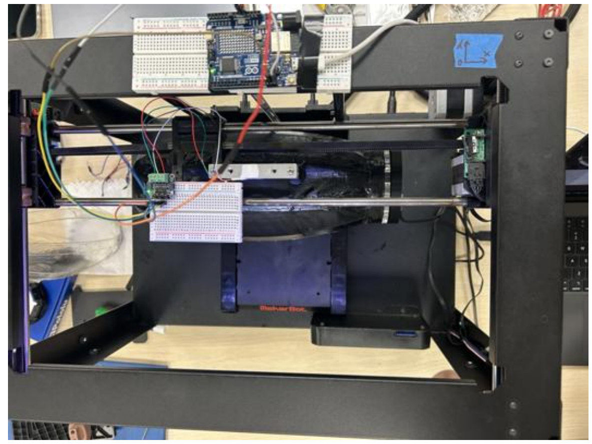
Spatiotemporal learning for tactile data. Li et al. [15] applied recurrent networks to tactile time series for slip detection. Sferrazza et al. [14] used CNNs with optical flow for force estimation from camera-based tactile sensors. ConvLSTM [13] has been widely adopted for spatiotemporal sequence prediction in video and radar applications, and is directly applicable to our wave propagation sensing task.

3. Hardware and System Design

The data collection rig repurposes a MakerBot Replicator 2 3D printer as a precision linear stage. The extruder assembly is replaced with a spherical indentor ($r = 5\,\text{mm}$) rigidly coupled to a SparkFun NAU7802 Qwiic load cell, which is sampled at 40 SPS via an Arduino Mega at 115200 baud. A USB fisheye camera (640$\times$480, 30 fps) is mounted internally, looking upward at the



(a) Front view: elastomer skin mounted in the MakerBot indentation frame with spherical indentor above.



(b) Top view: Arduino Mega, NAU7802 load cell breakout board, and synchronized sensor wiring.

Figure 1: Data collection hardware. The MakerBot 3D printer is repurposed as a precision Z-axis linear stage delivering controlled spherical indentation at 0.167 mm/s across a 0–10 mm depth range.

underside of the elastomer skin. The camera, load cell, and stage position are synchronized in software so every camera frame carries a timestamp-matched force and depth label.

The elastomer skin is a transparent silicone sheet carrying a dense maze-like fractal pattern printed on its inner surface. The skin is tensioned and mounted on a rigid oval frame (225 mm $\times$ 30 mm) fixed within the MakerBot build volume. The curved geometry means the indentor-to-surface angle varies with contact location; this is accounted for analytically in the label generation pipeline (Section 5).

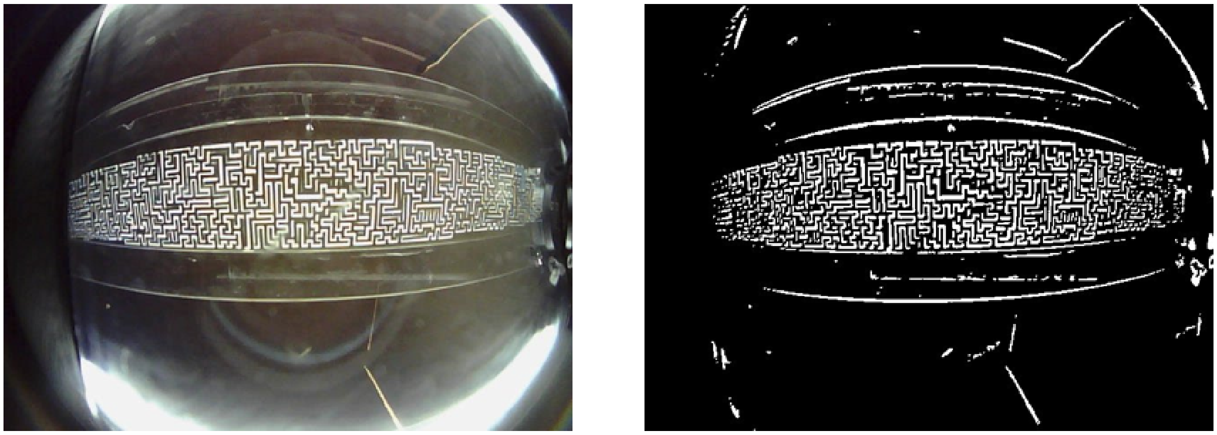
Table 1: Hardware components and specifications.

Component	**Model**	**Specification**
Linear stage	MakerBot Replicator 2	0.167 mm/s, 0–10 mm range
Camera	USB fisheye	640$\times$480 @ 30 fps
Load cell	SparkFun NAU7802	40 SPS, $\pm$5 N range
Indentor	Stainless steel sphere	$r = 5$ mm
Skin	Transparent silicone	Maze fractal pattern, 225$\times$30 mm

4. Pattern Design and Preprocessing

The sensing principle depends on reliably tracking the deformation of the printed maze pattern across frames. The pattern is designed to be high-spatial-frequency, aperiodic, and self-similar at multiple scales, so that both large (deep indentation) and subtle (wave propagation) deformations produce unique, trackable signatures.

Raw fisheye frames undergo a four-stage preprocessing pipeline before model input:



(a) Raw fisheye frame from the internal camera.

(b) Extracted binary pattern after the four-stage pipeline.

Figure 2: Pattern extraction. The maze-like fractal printed on the inner elastomer surface deforms under contact; the preprocessing pipeline isolates this deformation signal from ambient illumination variation.

1. **CLAHE** (Contrast-Limited Adaptive Histogram Equalization) — normalizes local contrast in 8×8 tiles, making the pipeline robust to illumination variation and vignetting from the fisheye lens.

2. **Adaptive thresholding** — binarizes the image using a local neighborhood comparison (block size 21, constant $C = 2$), separating pattern from background regardless of global brightness.

3. **Brightness floor** — rejects pixels below intensity 130 to suppress dim noise artifacts not removed by thresholding.

4. **Morphological opening** — applies a 3×3 structuring element to remove isolated single-pixel noise while preserving the connectivity of pattern lines.

Both the binary extracted pattern and the raw grayscale frame are retained as model inputs. The binary representation provides robust structural deformation information; the grayscale preserves amplitude information relevant to subtle wave-induced deformations that fall below the binary threshold.

5. Contact Mechanics and Label Generation

5.1 Hertzian Contact Model

For a spherical indentor of radius R pressed into a flat elastic half-space to depth δ, the classical Hertz contact theory [6] gives the contact radius a and normal force F:

$$a = \sqrt{R\delta}, \qquad F = \frac{4}{3}E^*\sqrt{R}\,\delta^{3/2} \tag{1}$$

where the reduced modulus E^* is defined as:

$$E^* = \frac{E}{1-\nu^2} \tag{2}$$

with Young's modulus E and Poisson's ratio ν. For the silicone elastomer skin, $\nu \approx 0.48$ (nearly incompressible). The pressure distribution within the contact patch follows the Hertzian profile:

$$p(r) = p_0\left(1 - \frac{r^2}{a^2}\right)^{1/2}, \qquad p_0 = \frac{3F}{2\pi a^2} \tag{3}$$

5.2 Surface Tension Correction

For soft elastomers with shear modulus $G < 100\,\mathrm{kPa}$, elastic surface tension γ (units: N/m) becomes non-negligible [9]. The elasto-capillary length $\ell_{ec} = \gamma/E$ sets the scale below which surface energy dominates bulk elasticity. The corrected contact force incorporating surface tension is:

$$F = \frac{4}{3}E^*\sqrt{R}\,\delta^{3/2} - 2\pi\gamma a \tag{4}$$

where the second term represents the adhesion-like pull from surface energy. We fit E and γ simultaneously to each per-strip force-displacement curve using nonlinear least squares minimization of:

$$\mathcal{L}(E,\gamma) = \sum_i \left(F_i^{\mathrm{meas}} - F(E,\gamma,\delta_i)\right)^2 \tag{5}$$

5.3 Full-Surface Deformation Field

Away from the contact patch, the surface displacement $u_z(r)$ follows the Boussinesq solution for a point load on an elastic half-space [10]:

$$u_z(r) = \frac{F(1-\nu^2)}{\pi E} \cdot \frac{1}{\max(r,\, a)} \cdot \left(1 - \frac{2\pi\gamma a}{F}\right) \tag{6}$$

where r is the distance from the contact center. This equation is evaluated at every point on the $2\,\mathrm{mm} \times 2\,\mathrm{mm}$ grid to generate the dense surface deformation labels used for training the spatial output heads.

5.4 Spatially Varying Calibration

Because the elastomer is mounted on a rigid curved frame, effective stiffness varies spatially. Near the boundary edges ($y \approx 0\,\text{mm}$ and $y \approx 18\,\text{mm}$), the rigid frame constrains lateral material flow, producing higher apparent stiffness. At the center ($y \approx 9\,\text{mm}$), the material deforms freely. We perform independent (E, γ) fits for each y-strip, producing a stiffness map $E(x, y)$, $\gamma(x, y)$ that is interpolated bilinearly to evaluate at arbitrary contact locations.

The surface stress tensor components at the contact point under axisymmetric loading are:

$$\sigma_{rr} = \frac{p_0}{2}\left[(1-2\nu)\frac{a^2}{r^2}\left(1-\left(1-\frac{r^2}{a^2}\right)^{3/2}\right)-\left(1-\frac{r^2}{a^2}\right)^{1/2}\right] \tag{7}$$

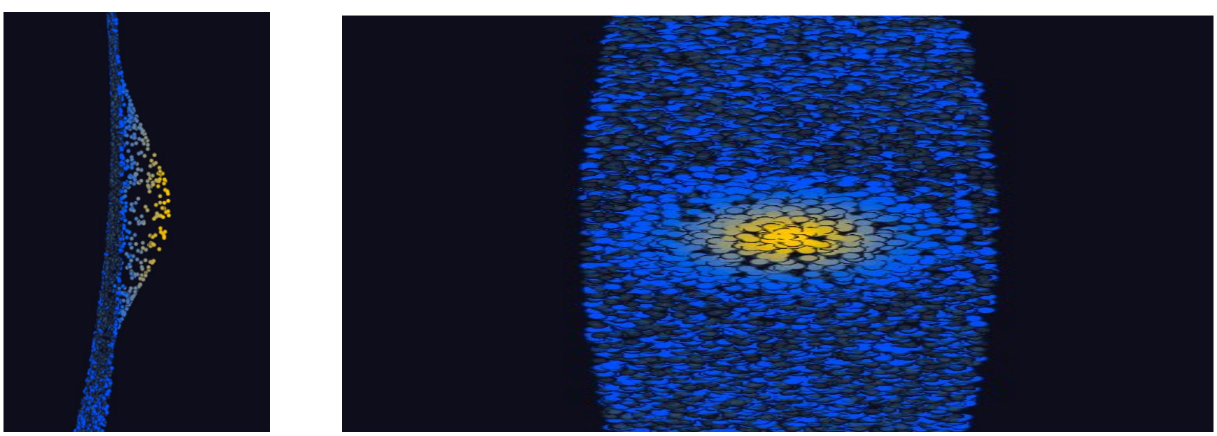
$$\sigma_{\theta\theta} = -\frac{p_0}{2}\left[(1-2\nu)\frac{a^2}{r^2}\left(1-\left(1-\frac{r^2}{a^2}\right)^{3/2}\right)+\left(1-\frac{r^2}{a^2}\right)^{1/2}\right] \tag{8}$$

The corresponding surface strain in the radial direction, relevant to the observed pattern displacement in the camera image, is:

$$\varepsilon_{rr} = \frac{1}{E}\left(\sigma_{rr} - \nu\,\sigma_{\theta\theta}\right) \tag{9}$$

These strain fields directly predict the expected pixel displacement of the fractal pattern at each grid location, providing a physics-based consistency check on the learned model output.

6. Dataset



(a) Side view.

(b) Top view. Yellow = maximum displacement at contact; blue = undeformed surface.

Figure 3: Generated point cloud labels for a representative indentation at the skin center. Displacement at every grid cell is computed from Eq. (6) using the locally calibrated material parameters.

The full dataset is collected on a uniform $2\,\text{mm} \times 2\,\text{mm}$ grid across the central $150\,\text{mm}$ of the oval skin surface. A $2\,\text{mm}$ grid spacing across a $150\,\text{mm} \times 30\,\text{mm}$ region yields **1,076 contact locations**. At each location, a full 0–10 mm indentation ramp is executed at 0.167 mm/s, captured at 30 fps, producing approximately 300 frames per location and **≈360,000 labeled frames** in total. The full perpendicular indentation dataset has been collected.

Each training sample consists of:

- A window of $N = 16$ consecutive camera frames (raw grayscale + binary pattern).
- Contact location (x, y) in mm, per-frame from the robot stage encoder.
- Indentation depth δ in mm, per-frame from the stage trajectory.
- Applied force F in N, per-frame from the NAU7802 load cell at 40 SPS.
- Surface deformation field $\mathbf{u}_z \in \mathbb{R}^{1076}$, computed from Eq. (6) with locally calibrated (E, γ).

Two sessions ($y = -6\,\text{mm}$ and $y = -14\,\text{mm}$) are held out entirely from training for validation. All labels are min-max normalized from training-set statistics only to prevent data leakage.

A complementary *drag dataset* is planned: the indentor swept at 5 angles (0°, 45°, 90°, 135°, 180°), 3 depths (2, 5, 8 mm), and 3 speeds (10, 30, 60 mm/s), producing ∼45 additional trajectories. These will be combined with the static dataset under identical schema, with weighted sampling (drag oversampled 10×) to prevent the larger static set from overwhelming the motion signal in training.

7. Model Architecture

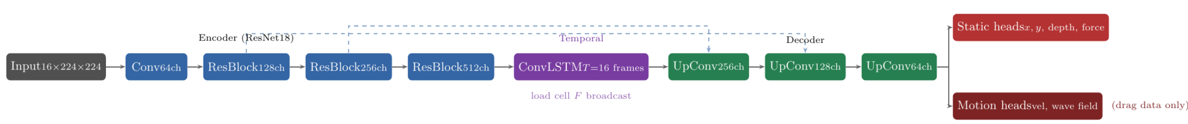


Figure 4: Encoder-decoder architecture. A ResNet18 encoder extracts spatial features from each frame; skip connections (dashed) preserve multi-scale structure through the decoder. A ConvLSTM processes the temporal sequence of feature maps. Two families of output heads are supervised independently via masked loss.

7.1 Spatial Encoder

Each input frame (concatenated grayscale + binary pattern + optical flow channels, $5 \times 224 \times 224$) is processed by a ResNet18 backbone [11] pre-trained on ImageNet and fine-tuned end-to-end. The backbone produces a spatial feature map $\mathbf{F} \in \mathbb{R}^{512 \times 7 \times 7}$ for each frame. Dense optical flow (u, v) between consecutive frame pairs is computed using the Farneback algorithm [12] and stacked as additional input channels, providing explicit motion information. The scalar load-cell force F_t is broadcast spatially to produce a force-conditioned feature tensor.

7.2 Temporal Model

A ConvLSTM [13] processes the sequence of $T = 16$ spatial feature maps, maintaining a hidden state $\mathbf{H}_t \in \mathbb{R}^{C\times H\times W}$ at every spatial location:

$$\mathbf{H}_t, \mathbf{C}_t = \mathrm{ConvLSTM}\big(\mathbf{F}_t,\, \mathbf{H}_{t-1},\, \mathbf{C}_{t-1}\big) \tag{10}$$

Unlike a standard LSTM operating on global feature vectors, the ConvLSTM preserves spatial structure through the recurrent computation, enabling the model to track spatially localized wave fronts and contact dynamics simultaneously across the skin surface.

7.3 Decoder and Output Heads

A symmetric decoder with three upsampling blocks (transposed convolutions with batch normalization and ReLU) and skip connections from the corresponding encoder scales recovers the spatial resolution of the input. The decoder output $\mathbf{D} \in \mathbb{R}^{64\times 224\times 224}$ feeds two families of output heads:

Static heads (supervised by all data):

- Contact location: $(\hat{x}, \hat{y}) \in \mathbb{R}^2$
- Indentation depth: $\hat{\delta} \in \mathbb{R}$
- Force map: $\hat{\mathbf{p}} \in \mathbb{R}^{1076}$ (N/mm^2 per grid cell)

Motion heads (drag data only, masked to zero loss on static samples):

- Contact velocity: $(\hat{v}_x, \hat{v}_y) \in \mathbb{R}^2$
- Surface wave field: $\hat{\mathbf{w}} \in \mathbb{R}^{1076\times 2}$ (wave direction and amplitude per cell)

7.4 Training Objective

The total loss is:

$$\mathcal{L} = \lambda_{\mathrm{loc}}\, \mathcal{L}_1(\hat{x}, x) + \lambda_\delta\, \mathcal{L}_1(\hat{\delta}, \delta) + \lambda_F\, \mathcal{L}_1(\hat{F}, F) + \nVdash_{\mathrm{drag}}\big[\lambda_v\, \mathcal{L}_1(\hat{\mathbf{v}}, \mathbf{v}) + \lambda_w\, \mathcal{L}_1(\hat{\mathbf{w}}, \mathbf{w})\big] \tag{11}$$

where $\nVdash_{\mathrm{drag}}$ is the masking indicator that zeros the motion head loss on perpendicular indentation samples. L_1 loss is used throughout for robustness to occasional force-measurement outliers. The model is optimized with Adam: learning rate 10^{-4} for the regression heads and 10^{-5} for the backbone, with cosine annealing over 150 epochs.

8. Experimental Results

8.1 Baseline Regressor

A baseline ResNet18 regressor (no temporal model, single-frame input) was trained on the initial 9-location dataset to validate the sensing pipeline before full dataset collection. Validation sessions at $y = -6\,\text{mm}$ and $y = -14\,\text{mm}$ were withheld entirely from training.

Table 2: Baseline single-frame ResNet18 performance on held-out sessions.

Output	Val MAE	Train MAE	Max error
Contact location y	0.52 mm	0.34 mm	1.8 mm
Indentation depth	0.38 mm	0.23 mm	1.2 mm
Force	55 mN	26 mN	180 mN

The model generalizes to unseen contact positions within ~0.5 mm despite the substantial variation in force range across the skin (0–2.38 N at $y = 0$, 0–0.50 N at $y = 8\,\text{mm}$). This confirms that the maze pattern deformation alone carries sufficient information to disambiguate both contact location and applied force.

8.2 Grad-CAM Interpretability

Gradient-weighted Class Activation Maps (Grad-CAM) [16] applied to the final convolutional layer confirm that the model attends to physically meaningful regions:

- **Location** (y): attention concentrates on the fringe of the contact zone at shallow depths, where lateral pattern displacement encodes position most clearly; at deep indentations, attention spreads across the full deformed region.
- **Depth**: activation maximizes at the contact center where pattern compression (inversely correlated with δ) is most pronounced.
- **Force**: saliency correlates with mid-field dot-density change, consistent with the Hertzian prediction that force depends on contact area rather than center displacement alone.

8.3 Full Dataset Training

The full 1,076-location dataset has been collected. Training the spatiotemporal encoder-decoder on the complete dataset and reporting quantitative results on 2D spatial generalization is ongoing work.

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